

Blowup Algebras

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1. What is a Blowup?

We care about studying varieties, and singularities are an obstruction to that. Intuitively, a singularity is a point of the variety where the tangent is not well defined. For instance, in $\mathbb{A}^2$ the curve $y^2 = x^3 + x^2$ has a node at the origin, where two branches cross and there are two perfectly valid tangent directions, $y = x$ and $y = -x$.

Blowups are a systematic tool for resolving singularities like this one. The idea is fairly straightforward, instead of picking one tangent direction at the singular point, we replace the point with a copy of $\mathbb{P}^1$, which parametrizes all lines through it simultaneously. The two branches then lift to the blowup as separate smooth curves, intersecting $\mathbb{P}^1$ at different points.

In general, the blowup of a variety $X \subset \mathbb{A}^n$ at a point replaces that point with a copy of $\mathbb{P}^{n-1}$, thus recording all the directions one can approach it from. So it lives in $\mathbb{A}^n \times \mathbb{P}^{n-1}$.

Projective space $\mathbb{P}^n$ parametrizes lines through the origin in $\mathbb{A}^{n+1}$, that is, a point $[p_0 : p_1 : \dots : p_n]$ represents the line through 0 and $(p_0, \dots, p_n)$, with two points identified if they differ by a nonzero scalar. Thus, a polynomial f only vanishes consistently at a projective point if $f(\lambda x) = 0$ whenever $f(x) = 0$, forcing f to be homogeneous. Subvarieties of $\mathbb{P}^n$ are thus cut out by homogeneous ideals. For the homogeneous coordinate ring $k[x_0, \dots, x_n]$ of $\mathbb{P}^n$, the projective Nullstellensatz correspondence sends a homogeneous radical ideal (other than the irrelevant ideal $(x_0, \dots, x_n)$, which gives $\emptyset$) to a projective subvariety; for example,

- (0) — all of $\mathbb{P}^n$ (as a set, $\mathbb{V}(0) = \mathbb{P}^n$);
- (x_i) — a hyperplane $\{x_i = 0\} \cong \mathbb{P}^{n-1}$;

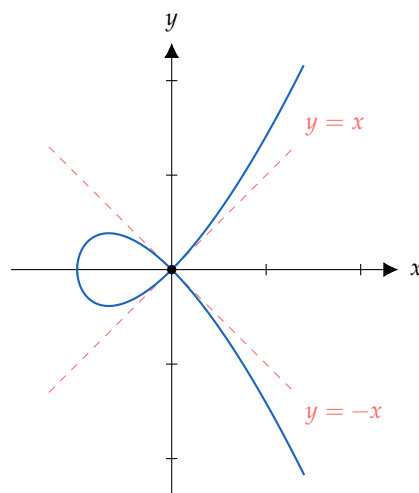


FIGURE 1. The nodal cubic $y^2 = x^3 + x^2$. The two branches cross at the origin with distinct tangent lines $y = x$ and $y = -x$.

- (x_i, x_j) — a codimension-2 linear subspace $\{x_i = x_j = 0\} \cong \mathbb{P}^{n-2}$;
- (f) for irreducible homogeneous f of degree d — a hypersurface of dimension $n - 1$.

2. Graded Rings and Modules

Since subvarieties of $\mathbb{P}^{n-1}$ are cut out by homogeneous ideals we need a framework to work with them. The polynomial ring $R = k[x_1, \dots, x_n]$ decomposes naturally by degree,

$$R = \bigoplus_{d \geq 0} k[x_1, \dots, x_n]_d$$

where $R_d = k[x_1, \dots, x_n]_d$ denotes homogeneous polynomials of degree d . Multiplication respects this decomposition, $R_d \cdot R_e \subseteq R_{d+e}$, and such an $R = \bigoplus_{d \geq 0} R_d$ is called a graded ring. The polynomial ring has a natural grading by degree, but a general ring does not. To extract a graded ring from R , we choose ideals $I_i \subseteq R$ forming a descending multiplicative chain,

$$R \supset I_1 \supset I_2 \supset I_3 \supset \dots$$

where $I_i I_j \subseteq I_{i+j}$. When the I_i are powers of a single ideal, this chain is called an I -adic filtration, and it allows us to measure “how deeply” an element lies in I . For our purposes, I is always the maximal ideal of a local Noetherian ring R .

From this filtration we can recover the associated graded ring $\text{gr}_I R$,

$$\text{gr}_I R := R/I \oplus I/I^2 \oplus I^2/I^3 \oplus \dots,$$

with a well-defined multiplication. For $a \in I^n/I^{n+1}$ and $b \in I^m/I^{m+1}$, we pick representatives $a' \in I^n$ and $b' \in I^m$ and define the product $ab \in I^{n+m}/I^{n+m+1}$ to be the image of $a'b'$. One can check that this is well-defined modulo I^{m+n+1} . We will now develop some theory around filtrations and graded rings that will be helpful later.

The idea of I -adic filtrations can be generalized to the case of modules.

DEFINITION 2.1. *Given an R -module M , and an ideal $I \subseteq R$ we get the I -adic filtration of the module,*

$$M \supset IM \supset I^2 M \supset \dots$$

and the associated graded $\text{gr}_I R$ -module,

$$\text{gr}_I M := M/IM \oplus IM/I^2 M \oplus I^2 M/I^3 M \oplus \dots$$

with multiplication defined as follows: for $a \in I^n/I^{n+1}$ and $b \in I^m M/I^{m+1} M$, we pick representatives $a' \in I^n$ and $b' \in I^m M$ and define the product $ab \in I^{n+m} M/I^{n+m+1} M$ to be the image of $a'b'$.

DEFINITION 2.2. *Let R be a ring, $I \subseteq R$ an ideal, and M an R -module. A filtration $M = M_0 \supset M_1 \supset \dots$ is called an I -filtration if $IM_n \subseteq M_{n+1}$ for all $n \geq 0$, and I -stable if there exists m such that $M_{m+i} = I^i M_m$ for all $i \geq 0$.*

REMARK 2.3. *Note that intersecting the terms of the I -adic filtration of M with some submodule $M' \subset M$ sadly generally does not give us the I -adic filtration of M' .*

The Artin-Rees lemma characterizes exactly when this does hold. We state it now and will prove it when the requisite theory is in place.

LEMMA 2.4 (Artin-Rees). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subset M$ be finitely generated R -modules. If*

$$M = M_0 \supset M_1 \supset M_2 \supset \cdots$$

is an I -stable filtration then the induced filtration

$$M' = M' \cap M_0 \supset M' \cap M_1 \supset M' \cap M_2 \supset \cdots$$

is also I -stable, i.e., there exists m such that $I(M' \cap M_n) = M' \cap M_{n+1}$ for $n \geq m$.

NOTATION. We write M_k for $I^k M$, and

$$\mathrm{gr}_I R = \bigoplus_{k \geq 0} I^k / I^{k+1}, \quad \mathrm{gr}_I M = \bigoplus_{k \geq 0} M_k / M_{k+1}$$

for the associated graded ring and module. We also need the *Rees algebra* (studied in detail in Section 3) and *Rees module*: for an I -filtration $\mathcal{J} : M = M_0 \supseteq M_1 \supseteq \cdots$,

$$B_I R := \bigoplus_{n \geq 0} I^n t^n \subseteq R[t], \quad B_{\mathcal{J}} M := \bigoplus_{n \geq 0} M_n t^n,$$

where $B_{\mathcal{J}} M$ is a graded $B_I R$ -module. (When $\mathcal{J}$ is the I -adic filtration, $M_n = I^n M$.)

There is no canonical ring homomorphism $R \rightarrow \mathrm{gr}_I R$ lifting the structure of R . The only canonical ring map out of R compatible with the grading is the projection onto degree 0,

$$R \twoheadrightarrow R/I = (\mathrm{gr}_I R)_0 \hookrightarrow \mathrm{gr}_I R,$$

which forgets everything in positive degree. The genuinely useful gadget attached to the associated graded is instead a *set* map (not a ring homomorphism), the initial form.

DEFINITION 2.5. *Let $f \in M$ and m be the greatest number such that $f \in M_m$. The initial form of f is*

$$\mathrm{in}(f) = f \pmod{M_{m+1}} \in M_m / M_{m+1} \subset \mathrm{gr}_I M$$

or if $f \in \cap^\infty M_m$

$$\mathrm{in}(f) = 0.$$

DEFINITION 2.6. *For a submodule $M' \subset M$ we define $\mathrm{in}(M')$ to be the $\mathrm{gr}_I R$ -submodule of $\mathrm{gr}_I M$ generated by $\mathrm{in}(f)$ for all $f \in M'$.*

REMARK 2.7. *The initial-form map is not additive, and it is not multiplicative when $\mathrm{gr}_I R$ has zero-divisors. Hence, unfortunately, $\mathrm{in}(M')$ is generally not obtained as the submodule generated by the initial forms of the generators of M' .*

For instance, take $R = k[x, y]$, $I = (x, y)$, and $J = (x^2 + y^3, xy) \subset R$ with the I -adic filtration. The initial forms of the generators are $\mathrm{in}(x^2 + y^3) = x^2$ and $\mathrm{in}(xy) = xy$, so the submodule generated by their initial forms is $(x^2, xy) \subset \mathrm{gr}_I R \cong k[x, y]$. However,

$$y \cdot (x^2 + y^3) - x \cdot (xy) = x^2 y + y^4 - x^2 y = y^4 \in J$$

so $\mathrm{in}(y^4) = y^4 \in \mathrm{in}(J)$. But $y^4 \notin (x^2, xy)$ in $k[x, y]$, since every element of (x^2, xy) has x as a factor while y^4 does not. Therefore $\mathrm{in}(J) \supsetneq (\mathrm{in}(x^2 + y^3), \mathrm{in}(xy))$.

PROPOSITION 2.8. *Let R be a ring, $I \subset R$ be an ideal, and let M be a finitely generated R -module with I -filtration $\mathcal{J} : M = M_0 \supset M_1 \supset \cdots$ by finitely generated modules M_i . The filtration $\mathcal{J}$ is I -stable if and only if the $B_I R$ -module $B_{\mathcal{J}} M$ is finitely generated.*

PROOF.

($\Rightarrow$)

Assume $\mathcal{J}$ is I -stable. Then we have $M_{n+i} = I^i M_n$ for some n and all $i \geq 0$. That is, for all $k > n$, M_k is generated from M_n by multiplying powers of I . Since each $M_0, \dots, M_n$ is finitely-generated, the union of their generating sets gives a finite generating set for $B_{\mathcal{J}} M$.

($\Leftarrow$)

Assume $B_{\mathcal{J}} M$ is finitely generated. Since $B_{\mathcal{J}} M = \bigoplus_{n \geq 0} M_n$ is a direct sum, every element has only finitely many nonzero homogeneous components. Therefore each generator g_i is contained in $M_0 \oplus \cdots \oplus M_{k_i}$ for some finite k_i , and taking $n = \max(k_1, \dots, k_r)$, all generators are contained in $M_0 \oplus \cdots \oplus M_n$.

Each generator g_i decomposes as a finite sum of homogeneous pieces, and the $B_I R$ -submodule generated by all these homogeneous pieces contains the $B_I R$ -submodule generated by $\{g_i\}$, since each g_i is a sum of its homogeneous pieces. Since $\{g_i\}$ generates $B_{\mathcal{J}} M$, the homogeneous pieces also generate $B_{\mathcal{J}} M$. Hence we may assume $B_{\mathcal{J}} M$ is generated by homogeneous elements in $M_0, \dots, M_n$.

Since all generators live in degrees $\leq n$, any element of M_{n+i} is produced by multiplying a generator $m \in M_k$ with $k \leq n$ by some $a \in I^{n+i-k}$. Since $\mathcal{J}$ is an I -filtration, $I^{n-k} m \in M_n$, so $a \cdot m = I^i \cdot (I^{n-k} m) \in I^i M_n$. Therefore $M_{n+i} \subseteq I^i M_n$. The reverse inclusion $I^i M_n \subseteq M_{n+i}$ follows immediately from $\mathcal{J}$ being an I -filtration. Hence $M_{n+i} = I^i M_n$ for all $i \geq 0$, hence $\mathcal{J}$ is stable. $\square$

LEMMA 2.9 (Artin-Rees). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subset M$ be finitely generated R -modules. If*

$$M = M_0 \supset M_1 \supset M_2 \supset \cdots$$

is an I -stable filtration then the induced filtration

$$M' = M' \cap M_0 \supset M' \cap M_1 \supset M' \cap M_2 \supset \cdots$$

is also I -stable, i.e., there exists m such that $I(M' \cap M_n) = M' \cap M_{n+1}$ for $n \geq m$.

PROOF. Let

$$\mathcal{J}' : M' = M'_0 \supset M'_1 := M' \cap M_1 \supset M'_2 := M' \cap M_2 \supset \cdots$$

be the induced filtration on M' . The module $B_{\mathcal{J}'} M' = \bigoplus_{n \geq 0} (M' \cap M_n)$ is naturally a graded $B_I R$ -submodule of $B_{\mathcal{J}} M$.

Since $\mathcal{J}$ is I -stable, $B_{\mathcal{J}} M$ is finitely generated as a $B_I R$ -module by Proposition 2.8. Since R is Noetherian and I is finitely generated, $B_I R = \bigoplus_{n \geq 0} I^n t^n$ is a finitely generated R -algebra and hence Noetherian by the Hilbert basis theorem. Therefore the submodule $B_{\mathcal{J}'} M'$ of the finitely generated $B_I R$ -module $B_{\mathcal{J}} M$ is itself finitely generated. Applying Proposition 2.8 again, $\mathcal{J}'$ is an I -stable filtration of M' . $\square$

REMARK 2.10. One of the important applications of associated graded constructions is when $I = \mathfrak{m}$ is the maximal ideal of a Noetherian local ring, $\text{gr}_{\mathfrak{m}} R$ is a finitely generated graded algebra over the field $R/\mathfrak{m}$, i.e. a quotient of a polynomial ring by a homogeneous ideal. Thus, gr_I allows us to turn arbitrary Noetherian rings into finitely-generated graded rings, which matters because such algebras are exactly the homogeneous coordinate rings of projective schemes, so passing to $\text{gr}_{\mathfrak{m}} R$ lets us deploy the full toolkit of projective geometry (Proj, Hilbert functions, Hilbert polynomials, dimension, multiplicity) on an object built from an otherwise unstructured local ring.

3. Blow Up Algebras

DEFINITION 3.1. Let R be a ring and $I \subseteq R$ an ideal. The **blowup algebra** (Rees algebra) of I in R is the R -algebra

$$B_I R := \bigoplus_{n \geq 0} I^n t^n = R \oplus It \oplus I^2 t^2 \oplus \cdots \cong R[tI] \subset R[t]$$

where t is a formal variable tracking the degree of the ideal. The associated graded ring is obtained by quotienting by $IB_I R$:

$$B_I R / IB_I R = \text{gr}_I R = R/I \oplus I/I^2 \oplus \cdots$$

The blowup algebra is the algebraic structure behind the geometric operation of blowing up a subvariety. Suppose R is the coordinate k -algebra of an affine variety X , and $I \subset R$ is the ideal of a subvariety $Y \subset X$. Let $a_1, \dots, a_r$ be k -algebra generators for R and let $g_0, \dots, g_s$ be generators of I as an ideal of R .

PROPOSITION 3.2. The blowup algebra $B_I R$ is a homomorphic image of $k[x_1, \dots, x_r, y_0, \dots, y_s]$.

PROOF. Define a ring homomorphism

$$\phi : k[x_1, \dots, x_r, y_0, \dots, y_s] \rightarrow B_I R, \quad x_i \mapsto a_i, \quad y_j \mapsto g_j t$$

This is surjective since any element of $B_I R$ is a finite sum of terms of the form $f \cdot g t^n$ with $f \in R$ and $g \in I^n$, and since f is a polynomial in the a_i and g is a polynomial in the g_j , every such term is in the image of ϕ . By the first isomorphism theorem:

$$B_I R \cong k[x_1, \dots, x_r, y_0, \dots, y_s] / \ker(\phi)$$

Since $B_I R$ is a graded ring and ϕ respects the grading by sending the x_i to degree 0 elements and the y_j to degree 1 elements, the kernel is a homogeneous ideal. $\square$

Here $k[x_1, \dots, x_r]$ is the coordinate ring of $\mathbb{A}^r$, and $k[y_0, \dots, y_s]$ is the homogeneous coordinate ring of $\mathbb{P}^s$. So $k[x_1, \dots, x_r, y_0, \dots, y_s]$ is the coordinate ring of the product $\mathbb{A}^r \times \mathbb{A}^{s+1}$, and quotienting by a homogeneous ideal in the y_j variables makes the vanishing scale-invariant in the y_j and hence cuts out a subvariety that is projective in the y -directions,

$$Z = \mathbb{V}(\ker \phi) \subset \mathbb{A}^r \times \mathbb{P}^s$$

The projection $\pi : \mathbb{A}^r \times \mathbb{P}^s \rightarrow \mathbb{A}^r$ restricts to a map $Z \rightarrow X$ which is an isomorphism away from the preimage of Y . The set Z is called the **blowup of Y in X** , and the preimage $\pi^{-1}(Y) \subset Z$ is called the **exceptional set**.

Algebraically, the exceptional set comes from killing I inside the blowup algebra: $B_I R / IB_I R = \text{gr}_I R$, so

$$\pi^{-1}(Y) = \text{Proj}(\text{gr}_I R) \subseteq \mathbb{P}^s.$$

Recall that $\text{gr}_I R = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is a graded ring, and its homogeneous primes not containing the irrelevant ideal $(\text{gr}_I R)_{>0}$ correspond to points of a projective variety, exactly as for $k[y_0, \dots, y_s]$ and $\mathbb{P}^s$. This variety — the projectivized normal cone of Y in X — parametrizes the limiting directions of approach to Y inside X : when Y is the origin in $X \subseteq \mathbb{A}^r$ it is the projectivized tangent cone, with one point for each limiting direction of a secant line to X through Y . We make this precise in Section 4.

We can now revisit the example from the start of the note more formally.

Blowing up the node of $y^2 = x^3 + x^2$.

The curve $X = \mathbb{V}(y^2 - x^3 - x^2)$ has a node at the origin. With $f = y^2 - x^3 - x^2$, the Jacobian $(\partial_x f, \partial_y f) = (-3x^2 - 2x, 2y)$ vanishes at $(0, 0)$, and $f(0, 0) = 0$, so the origin is singular.

We blow up $Y = \{0\}$ inside X . The coordinate ring is $R = k[x, y] / \langle y^2 - x^3 - x^2 \rangle$, and the maximal ideal of the origin is $I = \langle x, y \rangle$ (by Nullstellensatz).

Both R (as a k -algebra) and I (as an ideal) are generated by x, y . Relabeling the generators of R as x_1, x_2 and the generators of I as y_0, y_1 , the blowup algebra is a homomorphic image of $k[x_1, x_2, y_0, y_1]$ by the surjec-

$$\text{tion, } \phi : \begin{cases} x_1 \mapsto x, \\ x_2 \mapsto y, \\ y_0 \mapsto xt, \\ y_1 \mapsto yt \end{cases} \quad \text{by Proposition 3.2.}$$

Two relations are visibly in $\ker \phi$: $F = x_2^2 - x_1^3 - x_1^2$, and the $G = x_1 y_1 - x_2 y_0$, since $\phi(G) = x \cdot yt - y \cdot xt = 0$. Both are homogeneous of degree ≤ 1 in y_0, y_1 , so they cut out a subset $Z \subseteq \mathbb{A}^2 \times \mathbb{P}^1$.

To understand Z as a subset of $\mathbb{A}^2 \times \mathbb{P}^1$, we cover $\mathbb{P}^1$ by two pieces and examine Z over each. Recall a point of $\mathbb{P}^1$ is a class $[y_0 : y_1]$ under scalar identification. Every such point has either $y_0 \neq 0$ or $y_1 \neq 0$, so $\mathbb{P}^1$ is covered by the two subsets

$$U_0 = \{[y_0 : y_1] : y_0 \neq 0\}, \quad U_1 = \{[y_0 : y_1] : y_1 \neq 0\}.$$

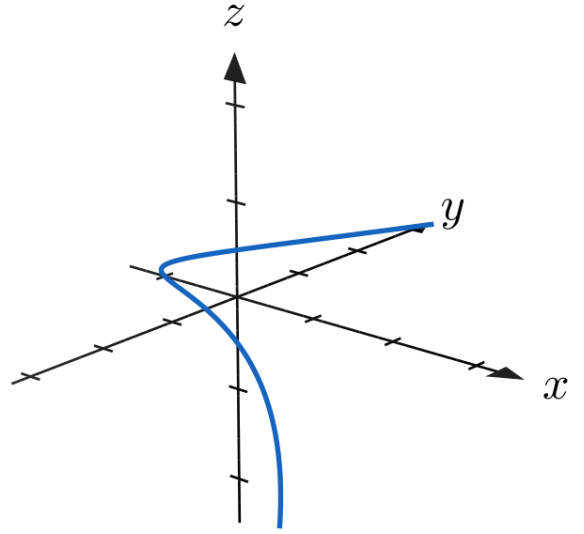


FIGURE 2. Resolving the node: the slope $z = y/x$ lifts the nodal cubic to the smooth space curve $(z^2 - 1, z^3 - z, z)$, whose two points over the origin ($z = \pm 1$) are identified under projection to the xy -plane.

On U_0 we may scale the representative so that $y_0 = 1$, leaving a single free coordinate $z := y_1/y_0$; this identifies U_0 with the affine line $\mathbb{A}^1$ with coordinate z . Likewise $U_1 \cong \mathbb{A}^1$ with coordinate $w := y_0/y_1$. We call U_0, U_1 the two **charts** of $\mathbb{P}^1$. Accordingly $\mathbb{A}^2 \times \mathbb{P}^1$ is covered by $\mathbb{A}^2 \times U_0$ and $\mathbb{A}^2 \times U_1$, with coordinates (x_1, x_2, z) and (x_1, x_2, w) respectively, and we study Z in each chart separately.

The chart U_0 (where $y_0 \neq 0$, coordinate $z = y_1/y_0$). Here we may set $y_0 = 1$ and $y_1 = z$. The relation $G = x_1y_1 - x_2y_0$ becomes

$$x_1z - x_2 = 0, \quad \text{i.e.} \quad x_2 = zx_1.$$

Geometrically this says exactly that $z = x_2/x_1$ is the slope y/x wherever $x_1 \neq 0$. Substituting $x_2 = zx_1$ into $F = x_2^2 - x_1^3 - x_1^2$ gives

$$(zx_1)^2 - x_1^3 - x_1^2 = x_1^2(z^2 - x_1 - 1) = 0.$$

Thus, inside the chart $\mathbb{A}^2 \times U_0$, the set Z is cut out by

$$x_2 = zx_1 \quad \text{and} \quad x_1^2(z^2 - x_1 - 1) = 0.$$

The equation $x_1^2(z^2 - x_1 - 1) = 0$ factors into two pieces:

- $x_1 = 0$: this lies entirely over the origin of $\mathbb{A}^2$ (since $x_1 = 0$ forces $x_2 = zx_1 = 0$), and is part of the exceptional set $\pi^{-1}(Y)$;
- $z^2 - x_1 - 1 = 0$: the **strict transform**, the closure of the part of Z lying over $X \setminus \{0\}$.

On the strict transform we may solve $x_1 = z^2 - 1$, and then $x_2 = zx_1 = z^3 - z$. Hence in this chart the strict transform is the image of the map

$$z \longmapsto (x_1, x_2, z) = (z^2 - 1, z^3 - z, z),$$

which is exactly the smooth space curve of Figure 2. It is smooth because it is the graph of a polynomial map in the single variable z .

The points of the strict transform lying over the origin $x_1 = x_2 = 0$ satisfy $z^2 - 1 = 0$, i.e. $z = \pm 1$. These are *two distinct points* of Z , with slopes $z = y/x = \pm 1$, corresponding to the two tangent directions $y = x$ and $y = -x$ of the original node. The blowup has pulled the two branches apart: they meet the chart's copy of $\mathbb{P}^1$ at the two different points $z = 1$ and $z = -1$.

The chart U_1 (where $y_1 \neq 0$, coordinate $w = y_0/y_1$). Here we set $y_1 = 1$ and $y_0 = w$. The relation $G = x_1y_1 - x_2y_0$ becomes

$$x_1 - x_2w = 0, \quad \text{i.e.} \quad x_1 = wx_2.$$

Substituting into $F = x_2^2 - x_1^3 - x_1^2$ gives

$$x_2^2 - (wx_2)^3 - (wx_2)^2 = x_2^2(1 - w^3x_2 - w^2) = 0.$$

Again this factors: $x_2 = 0$ lies over the origin (the exceptional set), while the strict transform is $1 - w^2 - w^3x_2 = 0$. The origin $x_1 = x_2 = 0$ on the strict transform forces $1 - w^2 = 0$, i.e. $w = \pm 1$, the same two tangent directions as before (note $w = x/y = 1/z$, consistent with $z = \pm 1$).

The two charts together cover all of Z , and on each the strict transform is smooth. Thus the blowup Z is smooth everywhere, and the single singular point of X has been replaced by the two points $z = \pm 1$ recording the two tangent lines of the node.

4. Tangent Cone

THEOREM 4.1 (Krull's Intersection Theorem). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely-generated R -module. There exists an element $r \in I$ such that*

$$(1 - r) \bigcap_{k=1}^{\infty} I^k M = 0$$

PROOF. Let $N = \bigcap_{k=1}^{\infty} I^k M$. We want to find $r \in I$ such that $(1 - r)N = 0$. Consider the I -adic filtration on M ,

$$M \supset IM \supset I^2 M \supset \dots$$

and the induced filtration on N ,

$$N \supset N \cap IM \supset N \cap I^2 M \supset \dots$$

By Artin-Rees applied to $N \subset M$, there exists m such that for all $n \geq m$,

$$N \cap I^n M = I^{n-m}(N \cap I^m M)$$

In particular,

$$N \cap I^{m+1} M = I(N \cap I^m M)$$

But since N is the intersection of all $I^k M$ we have $N \subseteq I^{m+1} M$, so $N \cap I^{m+1} M = N$. Similarly $N \subseteq I^m M$, so $N \cap I^m M = N$. Therefore:

$$N = IN$$

Since R is Noetherian and M is finitely generated, N is a finitely generated R -module. Applying Nakayama's lemma to $N = IN$: there exists $r \in I$ such that $(1 - r)N = 0$. $\square$

COROLLARY 4.1.1. *If R is a Noetherian domain or $(R, \mathfrak{m})$ is a local Noetherian ring, and $I \subsetneq R$ then*

$$\bigcap_{k=1}^{\infty} I^k = 0$$

PROOF. Let $M = R$. Since I is a proper ideal, there exists some $r \in I$ such that $r \neq 1$, so $1 - r \neq 0$. If R is a domain, we are done. If R is a local ring, $r \in I \subseteq \mathfrak{m}$ so $1 - r \notin \mathfrak{m}$, hence $1 - r$ is a unit and must be non-zero. Hence, $\bigcap_{k=1}^{\infty} I^k = 0$. $\square$

COROLLARY 4.1.2. *If R is a Noetherian local ring and $I \subsetneq R$ is an ideal then if $\text{gr}_I R$ is a domain, so is R .*

PROOF. If $fg = 0 \in R$ then $\text{in}(f)\text{in}(g) = 0 \in \text{gr}_I R$. Since $\text{gr}_I R$ is a domain, $\text{in}(f)$ or $\text{in}(g)$ is zero. Without loss of generality, assume $\text{in}(f) = 0$. Then by definition, $f \in \bigcap_{k=1}^{\infty} I^k$ and by KIT we know that $\bigcap_{k=1}^{\infty} I^k = 0$. Hence $f = 0$. Thus, R is a domain. $\square$

We now make the link between the exceptional set and the associated graded ring geometric. Throughout, $R = k[x_1, \dots, x_r]/J$ with $I = (x_1, \dots, x_r)$, k algebraically closed, and $J \subseteq I$ (so the origin lies on X); let $X = \mathbb{V}(J) \subseteq \mathbb{A}^r$.

The correspondence between the exceptional set in the blow up and the associated graded ring gives us a beautiful geometric consequence.

DEFINITION 4.2. *A line through 0 in $\mathbb{A}^r$ is determined by its direction, a point of $\mathbb{P}^{r-1}$: the line through 0 and $p = (p_1, \dots, p_r)$ corresponds to $[p_1 : \dots : p_r] \in \mathbb{P}^{r-1}$. A secant line to X through 0 has the form $\overline{0p}$ with $p \in X \setminus \{0\}$, giving a direction $[p] \in \mathbb{P}^{r-1}$, and a limiting secant direction is a limit point*

$$[p] = \lim_{n \rightarrow \infty} [p_n], \quad p_n \in X \setminus \{0\},$$

which exists after passing to a subsequence since $\mathbb{P}^{r-1}$ is compact. Let

$$D = \overline{\{[p] \in \mathbb{P}^{r-1} : p \in X \setminus \{0\}\}} \subseteq \mathbb{P}^{r-1}$$

denote the closure of the set of these directions. The (affine) tangent cone of X at 0 is the cone over D ,

$$TC_0(X) = \{0\} \cup \{x \in \mathbb{A}^r \setminus \{0\} : [x] \in D\} \subseteq \mathbb{A}^r,$$

the union of all lines through the origin in the limiting directions. Taking the closure is essential because it adds directions that arise as limits of secants but are not themselves the direction of any point of $X \setminus \{0\}$.

We define the tangent cone of X at 0 as the cone composed of all lines that are the limiting positions of secant lines to X passing through the point 0. Alternatively, the tangent cone at 0 is then the closure of the set of directions of points on X , together with the actual lines they represent in $\mathbb{A}^r$.

$$TC_0(X) = \overline{\{[p] \in \mathbb{P}^{r-1} : p \in X \setminus \{0\}\}} \subset \mathbb{P}^{r-1}$$

The closure captures limiting directions that arise as limits of secant lines but may not themselves be the direction of a point of $X \setminus \{0\}$.

DEFINITION 4.3. *The set of limiting secant directions,*

$$PTC_0(X) := \overline{\{[p] : p \in X \setminus \{0\}\}} \subseteq \mathbb{P}^{r-1},$$

is the projectivized tangent cone; it is the set D of Definition 4.2. Over $\mathbb{C}$ the closure is genuinely a set of limits: if $p_n \in X \setminus \{0\}$ with $p_n \rightarrow 0$, the directions $[p_n]$ have a convergent subsequence because $\mathbb{P}^{r-1}$ is compact. The affine tangent cone $TC_0(X) \subseteq \mathbb{A}^r$ is the cone over $PTC_0(X)$, and $PTC_0(X)$ is precisely the exceptional set of $\text{Bl}_0 X$ from Section 3.

THEOREM 4.4. *Let $R = k[x_1, \dots, x_r]/J$ with $I = (x_1, \dots, x_r)$ and k algebraically closed. The tangent cone $TC_0(X)$ to $X = \mathbb{V}(J) \subset \mathbb{A}^r$ at the origin is the algebraic subset:*

$$TC_0(X) = \mathbb{V}(\text{in}_I(J)) \subset \mathbb{A}^r$$

where $\text{in}_I(J) = \langle \text{in}(f) : f \in J \rangle \subset k[x_1, \dots, x_r]$ is the ideal generated by the initial forms of all elements of J . Moreover, the coordinate ring of the tangent cone is:

$$k[x_1, \dots, x_r]/\text{in}_I(J) \cong \text{gr}_I R$$

PROOF SKETCH. The theorem has two halves, algebraic and geometric, and the algebraic half does most of the work.

For the algebraic statement, write $S = k[x_1, \dots, x_r]$ and $\mathfrak{m} = (x_1, \dots, x_r) \subset S$, so $R = S/J$ and $I = \mathfrak{m}R$. The polynomial ring is its own associated graded for the $\mathfrak{m}$ -adic filtration: the quotient $\mathfrak{m}^n / \mathfrak{m}^{n+1}$ is naturally identified with the space S_n of degree- n forms (a polynomial in $\mathfrak{m}^n$ is remembered modulo $\mathfrak{m}^{n+1}$ only through its lowest, degree- n part), so $\text{gr}_{\mathfrak{m}} S \cong S$. Since $I^n = (\mathfrak{m}^n + J)/J$, the induced surjection $S \cong \text{gr}_{\mathfrak{m}} S \twoheadrightarrow \text{gr}_I R$ has kernel exactly the ideal generated by the initial forms $\text{in}_I(f)$, $f \in J$, namely $\text{in}_I(J)$. Hence

$$\text{gr}_I R \cong S / \text{in}_I(J) = k[x_1, \dots, x_r] / \text{in}_I(J).$$

For the geometric statement $TC_0(X) = \mathbb{V}(\text{in}_I(J))$, one inclusion is obvious: if $f \in J$, then $\text{in}_I(f)$ vanishes on every limiting secant direction, giving $TC_0(X) \subseteq \mathbb{V}(\text{in}_I(J))$. The reverse inclusion uses the fact that k is algebraically closed; we omit it, as a rigorous argument needs limiting tools beyond these notes. The worksheet walks through the rest of this proof in detail. $\square$

Revisiting the example we have been working with, for the nodal curve $y^2 = x^3 + x^2$, the lowest degree part of $f = y^2 - x^3 - x^2$ is $\text{in}_I(f) = y^2 - x^2 = (y - x)(y + x)$. So the tangent cone is $\mathbb{V}(y^2 - x^2) \subset \mathbb{A}^2$, the union of the two lines $y = x$ and $y = -x$, confirming the two tangent directions at the node. The coordinate ring is $k[x, y] / (y^2 - x^2) \cong \text{gr}_I R$.

We can now see the full picture for the node. In Section 1 we promised that blowing up replaces the singular point by a $\mathbb{P}^1$ recording tangent directions; in Section 3 we made this concrete, lifting the nodal cubic to a smooth curve whose two points $z = \pm 1$ over the origin record the two branches; and in this section we identified those directions purely algebraically. The three descriptions agree: the tangent cone $\mathbb{V}(\text{in}_I f) = \mathbb{V}(y^2 - x^2)$ is the pair of lines $y = \pm x$, its projectivization $\{[1:1], [1:-1]\} \subset \mathbb{P}^1$ is the exceptional set of the blowup, and its coordinate ring is $\text{gr}_I R$.

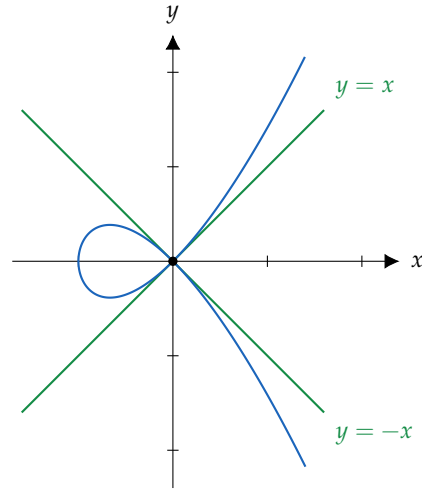


FIGURE 3. The tangent cone $TC_0(X) = \mathbb{V}(y^2 - x^2)$ is the pair of lines $y = \pm x$, matching the two branches' tangent directions and the two points $z = \pm 1$ found in the blowup of Section 3.

References and Credits

This note closely follows Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, and refers to Harris, *Algebraic Geometry: A First Course*. The illustrations are courtesy of Sair Shaikh '26.